Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/17/2005
PI Name:	Egolfopoulos, Fokion Ph.D.		
Project Title:	Quantitative Studies on the Propagation and Extinction of Near-Limit Premixed Flames Under Normal and Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	egolfopo@rcf.usc.edu	**Fax:**	FY 213 740-8071
PI Organization Type:	UNIVERSITY	**Phone:**	213 740-0480
Organization Name:	University of Southern California		
PI Address 1:	Department of Aerospace and Mechanical Engineering		
PI Address 2:			
PI Web Page:			
City:	Los Angeles	**State:**	CA
Zip Code:	90089-1453	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Miller, Fletcher J	**Contact Phone:**	619-594-5791
Contact Email:	fletcher.miller@sdsu.edu		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Spedding, Geoff (University of Southern California)		
Grant/Contract No.:	NNC04AA11A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	To develop a variable pressure, counterflow burner apparatus equipped with several diagnostic instruments - PIV, PSV, LDV, thermocouples, and high-speed cinematography, to study weakly burning laminar premixed and diffusion flames in both normal and microgravity conditions. To obtain the quantitative information about the characteristics of weakly burning flames in microgravity through the use of the 2.2s drop tower, and use the information to analyze flame near-limit phenomena. Additionally, to use/develop detailed 1- and 2-D numerical models to simulate flame structures, and to validate proposed kinetics. To obtain quantitative information about the characteristics of weakly burning flames in microgravity, and to use this information to analyze near-limit phenomena at normal gravity and microgravity conditions. The research will allow for the study of weakly burning flames without the complications of buoyancy, wall heat loss or time varying strain. It will provide fundamental data about the flammability limits, and has applications to turbulent combustion. Besides, proposed chemical kinetics can be validated and optimized in a wider range, and		

	subsequently better service the design of the next generation of internal combustion engines as well as air-breathing jet engines operating at speeds and altitudes that are noticeably greater compared to the ones currently achieved.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It will provide fundamental data about the flammability limits, and has applications to turbulent combustion.
Task Progress:	In FY04, our research effort was made on the following aspects: The rig has been re-organized and updated. Our experiments will include the determination of laminar flame speeds and extinction states for near-limit mixtures. For these measurements that DPIV technique will be employed. The variation of the mass flow rate is a very important aspect of the experimental procedure during the 2.2-sec drop time. To achieve flow rate adjustment a PID unit is employed and had to be integrate into the drop rig. In the previous design, the control circuit for the PID was not adjustable, but recently it was realized that, given the difference of flow rate change needed for different fuel/air mixtures and/or different equivalence ratio, the control circuit for the PID had to be adjustable. A new design of the control circuit has been finalized. The PID unit with the new adjustable control circuit is about to be tested. Additional improvements of the rig have been implemented. Near limit, ultra-lean H2/air premixed flames were systematically studied at normal gravity. A significant discrepancy between experimental and predicted extinction strain rates was identified. The possible reasons that could cause such a discrepancy were determined through the use of sensitivity analysis. As this mathematical tool is not readily available in the existing stagnation flow codes, a "brute force" approach was implemented for performing sensitivity analyses. The effect of uncertainty associated with both kinetics and molecular diffusion process on the extinction strain rate was systematically assessed. For the sensitivity on rate parameters, the pre-exponential factor, A, of each reaction was perturbed by ±25% and the Kext was subsequently determined. For the sensitivity on the mass diffusion coefficients, the mass diffusivity of each species i to the mixture, Di,m, perturbed by ±25% and the Kext was subsequently determined. In both cases, logarithmic sensitivity coefficients were formed by determining the derivatives d(lnKex)/d(lnA) and d(lnKext)/d(lnDi,m) respectively for kinetics and diffusion. As expected, the sensitivity analysis showed that the extinction strain rate is positively sensitive to the reactions H2 + O = OH + H , H2 + OH = H2O + H , HO2 + H = OH + OH , and the radical chain branching reaction H + O2 = OH + O. As also expected, the reaction H + O2 (+M) = HO2 (+M) exhibits a large negative sensitivity along with other radical-termination reactions, HO2 + OH = O2 + H2O HO2 + H = H2 + O2. It is reasonable to attribute a part of the observed discrepancies to uncertainties associated with the rate parameters of the aforementioned reactions. However, the influence of kinetic uncertainties cannot be adequately assessed without considering the uncertainties in the diffusion coefficients. The sensitivity analysis showed the diffusivity of four species had a notable effect on Kext, namely H2, O2, H2O, and H. More interesting, the logarithmic sensitivity coefficients for diffusion are not only comparable to the ones of reaction rates, but in the case of the H2 are nearly 3 times larger that those of reaction rates. Additionally, the logarithmic sensitivities on the diffusivities of other species such as H and O2 while smaller than that of H2 are still of the order of those of reaction rates. Obviously, the discrepancies between experiment and simulation cannot be conclusively viewed as the kinetics being deficient, because the diffusion uncertainties may be solely responsible for the discrepancies. That validating kinetics mechanism against Kext's without considering uncertainties related to diffusivities could result in the falsification of rate constants. It is important that the validation of models against flame data of such highly diffusive mixtures must be done by considering the uncertainties associated with both kinetics and molecular diffusion.
Bibliography Type:	Description: (Last Updated: 10/19/2009)
Articles in Peer-reviewed Journals	Dong Y, Holley AT, Andac MG, Egolfopoulos FN, Davis SG, Middha P, Wang H. "Premixed Extinction of H2/Air Flames: Chemical Kinetics and Diffusion Effects." Submitted to Combustion and Flame, November 2004. , Nov-2004
Presentation	Dong,Y.; Holley,A.T.; Andac,M.G.; Egolfopoulos,F.N.; Davis,S.G.; and Wang,H. "Premixed Extinction of H2/Air Flames: Chemical Kinetics and Diffusion Effects" WSS/CI 2004 Spring Meeting, UC Davis, USA Mar-2004